Supplemental Material for
"Wigner Resolution from Action Capacity"

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S1. A DE GOSSON–ZUREK–MULLOY DICTIONARY

The Letter fixes one notation. The same phase-space geometry appears in Zurek's account of sub-Planck structure [1] and in de Gosson's symplectic treatment of indeterminacy [2–7], each under different symbols and different normalizations. This section maps the Letter's symbols onto theirs. The Letter sets the entries. Zurek and de Gosson reach well past the Letter, into decoherence, chaos, Donoho–Stark and Hardy inequalities, Mahler volume, and the John ellipsoid. Only the part of each work that the Letter's symbols name is recorded here.

S1.1. Conventions and the convention bridge

The three works measure the same widths with three conventions. The Letter writes the covariance semi-axes as

$$\Delta x = \sqrt{2}\,\sigma_x, \qquad \Delta p = \sqrt{2}\,\sigma_p, \tag{S1}$$

so that the Heisenberg cell is the state's covariance ellipse and its area equals the symplectic capacity. Zurek writes the widths as standard deviations, $L \simeq \sigma_x$ and $P \simeq \sigma_p$, defined from $L^2 \simeq \langle x^2\rangle - \langle x\rangle^2$ and $P^2 \simeq \langle p^2\rangle - \langle p\rangle^2$. The two widths differ by a fixed factor,

$$\Delta x = \sqrt{2}\,L, \qquad \Delta p = \sqrt{2}\,P. \tag{S2}$$

The Letter carries the area factor π inside every cell, so the quantum of action reads $h/2 = \pi\hbar$. Zurek forms bare products, $A \simeq L \times P$, without the π. de Gosson measures a cell by its symplectic capacity c, normalized so that a quantum blob has $c = \pi\hbar = h/2$. Translating the action capacity across the three,

$$A \;=\; \pi\,\Delta x\,\Delta p \;=\; c(\Omega_\Sigma) \;=\; 2\pi\,\sigma_x\sigma_p \;=\; 2\pi\,(L \times P), \tag{S3}$$

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where Ω_Σ is de Gosson's covariance ellipsoid and $L \times P$ is Zurek's action.

One symbol collides across the three works and is worth flagging. In the Letter X and P are the scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$, and X_θ, P_θ are the rotated quadratures. In Zurek P is the momentum width and L the position width. In de Gosson $X \subset \mathbb{R}^n_x$ and $P \subset \mathbb{R}^n_p$ are convex bodies, the position and momentum data clouds. The dictionary below keeps each work in its own column so the three uses do not mix.

Zurek fixes these widths only to order of magnitude. He identifies the support widths L and P with the effective extent of the state, $L^2 \approx \langle x^2 \rangle - \langle x \rangle^2$, forms the bare action $A \approx L \times P$ without the area factor π, and writes the sub-Planck scale as $a \approx \hbar^2/A$. His interference relations $\delta_x = \hbar/P$, $\delta_p = \hbar/L$, and his tile area, Eqs. (7), (8), and (12), are exact once L and P are set, yet inherit the order-unity looseness of the support widths. In the Letter's covariance variables the same relations hold up to the fixed factors of Eqs. (S2) and (S3), $\sqrt{2}$ between the widths and 2π between the actions. The Letter writes $\sim$ for this reason and reserves equality for the relations symplectic polar duality fixes exactly.

S1.2. The dictionary

Tables S1 and S2 pair each symbol of the Letter with its counterpart in the two source works. The prose entries of Sec. S1 S1.3 expand the rows that carry the Letter's central relations, writing each relation out in all three notations.

S1.3. Entry by entry

a. Wigner function. The three works share one object. The Letter writes $W(x,p) = \frac{1}{\pi\hbar}\int \psi^*(x+s)\psi(x-s)\, e^{2ips/\hbar}\, ds$. Zurek's Eq. (2) is the same transform with the integration variable $y = 2s$ and the prefactor $1/2\pi\hbar$. de Gosson's Eq. (33) is the n-dimensional form $W\psi(z)$ on $\mathbb{R}^{2n}$. All three integrate to one and return the marginals $|\psi(x)|^2$ and $|\widehat{\psi}(p)|^2$.

b. Capacity semi-axes and the action capacity. The Letter's Heisenberg cell is $A = \pi\, \Delta x\, \Delta p \geq h/2$. Zurek's classical action is the bare product $A \simeq L \times P$ of the support widths, Eq. (4). de Gosson's covariance ellipsoid Ω_Σ carries the same content as a symplectic capacity. His Theorem 14 states that the quantum condition $\Sigma + \frac{i\hbar}{2} J \geq 0$ holds if and only if $c(\Omega_\Sigma) \geq \pi\hbar$ for every symplectic capacity c, which is if and only if Ω_Σ contains a quantum blob. The Letter's $A \geq h/2$ is this capacity bound, since $h/2 = \pi\hbar$ and $c(\Omega_\Sigma) = \pi\, \Delta x\, \Delta p = A$. The conversion to Zurek's product is

This Letter (Mulloy)	**Zurek [1]**	**de Gosson [2–4]**
Position, momentum $x,\ p$	$x,\ p$	$x \in \mathbb{R}^n_x,\ p \in \mathbb{R}^n_p$
Wigner function $W(x,p)$	$W(x,p)$, Eq. (2)	$W\psi(z)$, Eq. (33)
Capacity semi-axis (position) $\Delta x = \sqrt{2}\,\sigma_x$	L, with $L^2 \simeq \langle x^2\rangle - \langle x\rangle^2$	half-width of $X = \Pi_x\Omega_\Sigma$
Capacity semi-axis (momentum) $\Delta p = \sqrt{2}\,\sigma_p$	P, with $P^2 \simeq \langle p^2\rangle - \langle p\rangle^2$	half-width of $P = \Pi_p\Omega_\Sigma$
Heisenberg cell, action capacity $A = \pi\Delta x\Delta p \geq h/2$	classical action $A \simeq L \times P$, Eq. (4)	covariance ellipsoid Ω_Σ, $c(\Omega_\Sigma) \geq \pi\hbar$, Thm. 14
Resolution semi-axis (position) $\delta x = \hbar/\Delta p$	$\delta_x = \hbar/P \simeq \sqrt{2}\,\delta x$, Eq. (7)	$B_P(\Delta p)^\hbar = B_X(\hbar/\Delta p)$, Eq. (7)
Resolution semi-axis (momentum) $\delta p = \hbar/\Delta x$	$\delta_p = \hbar/L \simeq \sqrt{2}\,\delta p$, Eq. (8)	$B_X(\Delta x)^\hbar = B_P(\hbar/\Delta x)$, Eq. (7)
Quantum blob a_θ, action $h/2$	minimum-uncertainty Gaussian	$QB(S) = S\Big(B^{2n}(\sqrt{\hbar})\Big),\ c = \pi\hbar$, Eq. (26)
Principal blobs $a_{\theta=0},\ a_{\pi/2}$	axis interference scales δ_x, δ_p	diagonal case $S = M_L$; John$(X \times X^\hbar)$, Thms. 10, 12

TABLE S1. Dictionary, part 1: core geometry. The Letter's symbols and their counterparts in Zurek [1] and de Gosson [2–4]. Equation and theorem numbers in the Zurek column refer to [1]; those in the de Gosson column refer to [2]. Convention factors relating the columns are collected in Sec. S1 S1.1, Eqs. (S2)–(S3). Because Zurek measures with σ and the Letter with $\sqrt{2}\,\sigma$, his resolution scales δ_x, δ_p exceed the Letter's $\delta x, \delta p$ by the fixed factor $\sqrt{2}$. In de Gosson's column $B_X(R)$ and $B_P(R)$ are balls of radius R in position and momentum space and $X^\hbar$ is the $\hbar$-polar dual of X.

Eq. (S3), $A = 2\pi\,(L \times P)$.

c. Resolution scales and ħ-polar duality. This is the relation the dictionary turns on. The Letter writes

$$\delta x = \frac{\hbar}{\Delta p}, \qquad \delta p = \frac{\hbar}{\Delta x}. \tag{S4}$$

Zurek reaches the same scales for the smallest structures of the Wigner function, $\delta_x = \hbar/P$ in his Eq. (7) and $\delta_p = \hbar/L$ in his Eq. (8). de Gosson reaches them as $\hbar$-polar duality. For an interval of

This Letter (Mulloy)	Zurek [1]	de Gosson [2–4]
Quorum cell $\tilde{a} = \pi\delta x\delta p = \pi\hbar^2/(\Delta x\Delta p)$	sub-Planck action $a \simeq \hbar^2/A$, Eq. (3); tile $(2\pi\hbar)^2/A$, Eq. (12)	$\hbar$-polar dual $A^\hbar$; saturated pair $X \times X^\hbar$
Reciprocity $\tilde{a}\, A = (h/2)^2$, chain $A \supseteq a_\theta \supseteq \tilde{a}$	$a \simeq \hbar/N$, $N = A/2\pi\hbar$	biduality $(X^\hbar)^\hbar = X$, antimonotone, Eqs. (4),(5)
Quantum kernel K_θ (Wigner function of a_θ)	Husimi smoothing erases a (Methods)	$W\psi_{W,Y} = (\pi\hbar)^{-n} e^{-Gz\cdot z/\hbar}$, Eq. (41)
Convolution $\widetilde{W}_\theta = W * K_\theta$	not used	smoothing toward a non-negative distribution [5, 8]
Portrait $\widetilde{W} = \frac{1}{\pi}\int_0^\pi \widetilde{W}_\theta\, d\theta \geq 0$	not used	not used
Quadrature rotation X_θ, P_θ, angle θ	N, S, E, W of the compass	symplectic rotation $R \in Sp(n) \cap O(2n)$; covariance $W(\hat{S}^{-1}\psi) = W\psi \circ S$, Eq. (34)
Scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$; rigid ellipse $\hat{a}_\theta$	not used	normalization to $B^{2n}(\sqrt{\hbar})$ via M_L
Saturation $A = h/2$ (vacuum), $a_\theta = \tilde{a} = A$	smallest scale $a \to \hbar$	unique blob, $c = \pi\hbar$, saturated $X^\hbar = P$, Thm. 14

TABLE S2. Dictionary, part 2: derived and dynamical quantities. Conventions and column sources are as in Table S1.

half-width R the dual is the interval of half-width $\hbar/R$, his Lemma 1, Eq. (7),

$$B_P(\Delta p)^\hbar = B_X\left(\frac{\hbar}{\Delta p}\right) = B_X(\delta x), \qquad B_X(\Delta x)^\hbar = B_P\left(\frac{\hbar}{\Delta x}\right) = B_P(\delta p). \tag{S5}$$

The three statements coincide once the widths are converted by Eq. (S2). Because Zurek measures with σ and the Letter measures with $\sqrt{2}\,\sigma$, the Letter's resolution is finer than Zurek's nominal scale by the same fixed factor, $\delta x = \delta_x/\sqrt{2}$.

d. Quantum blob. The Letter's a_θ is a de Gosson quantum blob of action $h/2$, deforming at fixed action across θ. de Gosson's definition is $QB(S) = S(B^{2n}(\sqrt{\hbar}))$ for $S \in Sp(n)$, his Eq. (26), with volume $\frac{1}{n!}(h/2)^n$ and symplectic capacity $c = \pi\hbar$. In one freedom this is an ellipse of area $\pi\hbar = h/2$, the Letter's a_θ. Zurek's building block is the round minimum-uncertainty Gaussian, the coherent-state cell of area of order $\hbar$ from which his cat and compass states are assembled.

e. Principal blobs. The Letter's two principal members are $a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. In de Gosson these are the diagonal, axis-aligned case. His projection theorem, Theorem 10, gives a saturated polar pair, $(\Pi_X QB(S))^\hbar = \Pi_P QB(S)$, exactly when $S = M_L$, and his Theorem 12 identifies the unique blob inscribed in $X \times X^\hbar$ as the John ellipsoid. Zurek's interference scales δ_x, δ_p are these principal members read off the coordinate axes.

f. Quorum cell, polar dual, reciprocity. The Letter's quorum cell is $\tilde{a} = \pi\,\delta x\,\delta p = \pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the Heisenberg cell, with the reciprocity $\tilde{a}\,A = (h/2)^2$ and the chain $A \supseteq a_\theta \supseteq \tilde{a}$. de Gosson's $\hbar$-polar dual $A^\hbar$ is the same object, with biduality $(X^\hbar)^\hbar = X$ and antimonotonicity, his Eqs. (4) and (5). The reciprocity follows from the interval duals of Eq. (S5), since $\tilde{a}\,A = \pi^2(\delta x\,\Delta p)(\delta p\,\Delta x) = \pi^2\hbar^2 = (\pi\hbar)^2$. Zurek's sub-Planck action is $a \simeq \hbar^2/A$, his Eq. (3), with the compass tile $a = (2\pi\hbar)^2/A$ of his Eq. (12). In Zurek's widths the Letter's quorum cell is $\tilde{a} = (\pi/2)\,\hbar^2/(L \times P)$. He also records the count $N = A/2\pi\hbar$ of orthogonal states and the scale $a \simeq \hbar/N$.

g. Quantum kernel, convolution, portrait. The Letter's quantum kernel K_θ is the Wigner function of the blob a_θ, a centered Gaussian of action $h/2$. de Gosson's Eq. (41) gives the Wigner function of a generalized Gaussian, $W\psi_{W,Y}(z) = (\pi\hbar)^{-n} e^{-Gz\cdot z/\hbar}$, which is the kernel written through its covariance. Convolving W with a Gaussian of action $h/2$ returns a non-negative distribution, the route through Cartwright [8] and de Gosson's Husimi–Wigner cells [5]. The Letter's per-angle convolution is $\widetilde{W}_\theta = W * K_\theta$ and the portrait is the family integral $\widetilde{W} = \frac{1}{\pi}\int_0^\pi \widetilde{W}_\theta\,d\theta \geq 0$. Zurek notes the complementary fact that the Husimi function, the Wigner function smoothed on a full Planck scale, erases the sub-Planck scale a. The portrait sits between the two, resolved at $\tilde{a}$ rather than at h.

h. Quadrature rotation. The Letter's rotated quadratures are $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$, the homodyne quadratures at phase θ that orient the blob and the kernel. de Gosson carries the same rotation as a symplectic rotation $R \in Sp(n) \cap O(2n)$ and as the metaplectic covariance of the Wigner function, $W(\hat{S}^{-1}\psi) = W\psi \circ S$, his Eq. (34). The family integral over θ is the tomographic sweep [9, 10]. Zurek's compass orients its lobes north, south, east, and west, fixing the two axis directions of the interference grid.

S2. THE RESOLUTION OF THE PORTRAIT

The portrait is a single convolution. Because the convolution with K_θ and the integral over θ are both linear, they commute, and the family integral collapses to one kernel,

$$\widetilde{W} = \frac{1}{\pi}\int_0^\pi W * K_\theta \, d\theta = W * \bar{K}, \qquad \bar{K} = \frac{1}{\pi}\int_0^\pi K_\theta \, d\theta. \tag{S6}$$

The resolution of the portrait is a property of the one kernel $\bar{K}$. Non-negativity is the per-angle property of Sec. S3 S3.3, where each $W * K_\theta \geq 0$. Resolution is the property of $\bar{K}$ derived here. The two statements read the same object through its two ends, the family for non-negativity and the average for resolution.

S2.1. The chord function of the averaged kernel

In the chord domain the convolution is a product, so the chord function of the portrait is the chord function of W times the chord function of $\bar{K}$. Write $\widehat{f}$ for the symplectic Fourier transform of f and ξ for the chord, the conjugate of the scaled phase-space point. The chord function $\widehat{\bar{K}}$ multiplies $\widehat{W}$ chord by chord, so its reach in ξ sets the finest structure the portrait keeps.

The Letter writes each kernel through the capacity A,

$$K_\theta(x,p) \propto \exp\left[-X_\theta^2 - \frac{A^2}{(\pi\hbar)^2}\, P_\theta^2\right]. \tag{S7}$$

Polar duality carries the capacity to the quorum cell. With $A = (\pi\hbar)^2/\tilde{a}$ from the reciprocity $\tilde{a}\,A = (\pi\hbar)^2$, the narrow axis of every blob is the quorum semi-axis, and the kernel wears the resolution it produces,

$$\frac{A^2}{(\pi\hbar)^2} = \frac{(\pi\hbar)^2}{\tilde{a}^2}, \qquad K_\theta(x,p) \propto \exp\left[-X_\theta^2 - \frac{(\pi\hbar)^2}{\tilde{a}^2}\, P_\theta^2\right]. \tag{S8}$$

The Gaussian is broad along X_θ and narrow along P_θ by the factor $\tilde{a}/(\pi\hbar)$. Its chord function is broad where the kernel is narrow,

$$\widehat{K_\theta}(\xi) \propto \exp\left[-\tfrac{1}{4}\left(\xi_{X\theta}^2 + \frac{\tilde{a}^2}{(\pi\hbar)^2}\,\xi_{P\theta}^2\right)\right], \tag{S9}$$

reaching chords of order $\pi\hbar/\tilde{a}$ along the $\xi_{P\theta}$ axis. As θ runs over $[0,\pi)$ this long axis sweeps every direction.

The average over the family is the chord function of $\bar{K}$. Writing $|\xi|^2 = \xi_X^2 + \xi_P^2$ and carrying out the angular integral,

$$\widehat{\bar{K}}(|\xi|) = \exp\left[-\frac{|\xi|^2}{8}\left(1 + \frac{\tilde{a}^2}{(\pi\hbar)^2}\right)\right] I_0\left(\frac{|\xi|^2}{8}\left(1 - \frac{\tilde{a}^2}{(\pi\hbar)^2}\right)\right), \tag{S10}$$

with I_0 the modified Bessel function. The chord function depends on $|\xi|$ alone, so the average is isotropic in the scaled frame, and it equals one at zero chord, so the portrait keeps the bulk of W in place.

S2.2. The cutoff is the quorum cell

At large chord the Bessel function grows as $I_0(z) \sim e^z/\sqrt{2\pi z}$, and the two exponentials combine to leave the envelope

$$\widehat{\bar{K}}(|\xi|) \sim \frac{2}{\sqrt{\pi\,(1-\tilde{a}^2/(\pi\hbar)^2)}}\,\frac{1}{|\xi|}\,\exp\!\left[-\frac{\tilde{a}^2}{4(\pi\hbar)^2}\,|\xi|^2\right], \qquad |\xi| \gg 1. \tag{S11}$$

The Gaussian envelope is set by the quorum cell $\tilde{a}$ and not by the broad axis, so the average passes structure out to the chord

$$|\xi|_{\rm cut} \sim \frac{2\pi\hbar}{\tilde{a}}. \tag{S12}$$

Restoring physical units, the same envelope is the ellipse of semi-axes δx and δp,

$$\exp\!\left[-\tfrac{1}{4}\big(\delta x^2\,\xi_x^2 + \delta p^2\,\xi_p^2\big)\right], \qquad \delta x = \frac{\hbar}{\Delta p}, \quad \delta p = \frac{\hbar}{\Delta x}, \tag{S13}$$

the quorum cell $\tilde{a}$ written in the chord domain. The portrait holds the structure of W down to $\tilde{a}$ and removes structure finer than $\tilde{a}$. The factor $1/|\xi|$ ahead of the Gaussian softens the edge, so the cutoff is a gradual rolloff at $\tilde{a}$ rather than a sharp wall.

S2.3. Extent is the capacity, resolution is the quorum

The averaged kernel is broad in space and sharp in feature at once, and the two are different measures. The second moment of $\bar{K}$ averages the rotating blob over all angles and returns an isotropic spread set by the broad axis, the capacity scale A. The chord cutoff of Eq. (S12) returns the resolution set by the narrow axis, the quorum $\tilde{a}$. Extent and resolution are the major and the minor axes of the same blobs, the capacity and its polar dual. A width in space reports the capacity. The reach in chord reports the resolution. The portrait is wide as A and fine as $\tilde{a}$, the two cells the duality pairs.

This is the synthesis the Letter records. The kernel carries only the capacity A of the state, the axis along which the blob is broad. The average over the family resolves the portrait at the quorum cell $\tilde{a}$, the axis along which the blob is narrow. Zurek's sub-Planck scale $a \simeq \hbar^2/A$ [1] is this resolution up to the convention factors of Sec. S1 S1.1, and de Gosson's polar duality $\tilde{a}\,A =$

$(h/2)^2$ [3] is the step that carries the one cell to the other. Section S3 confirms that the portrait carries the structure of W at these chords and displaces no feature.

S3. NUMERICAL VALIDATION

The Letter states that the bare Wigner function W and the non-negative portrait $\widetilde{W}$ carry the same structure at the same positions, and that $\widetilde{W}$ is non-negative. This section measures both across the twelve states of the Letter. Every instrument runs identically on W and on $\widetilde{W}$ and reports position, the quantity the portrait preserves. No amplitude scale appears, so there is no ambiguity about what each number measures. The one asymmetry between the two functions is non-negativity, the defining property of the portrait. The resolution of that shared structure is the quorum cell $\tilde{a}$, derived in Sec. S2 and confirmed here as the transfer function of Eq. (S10) measured on every state.

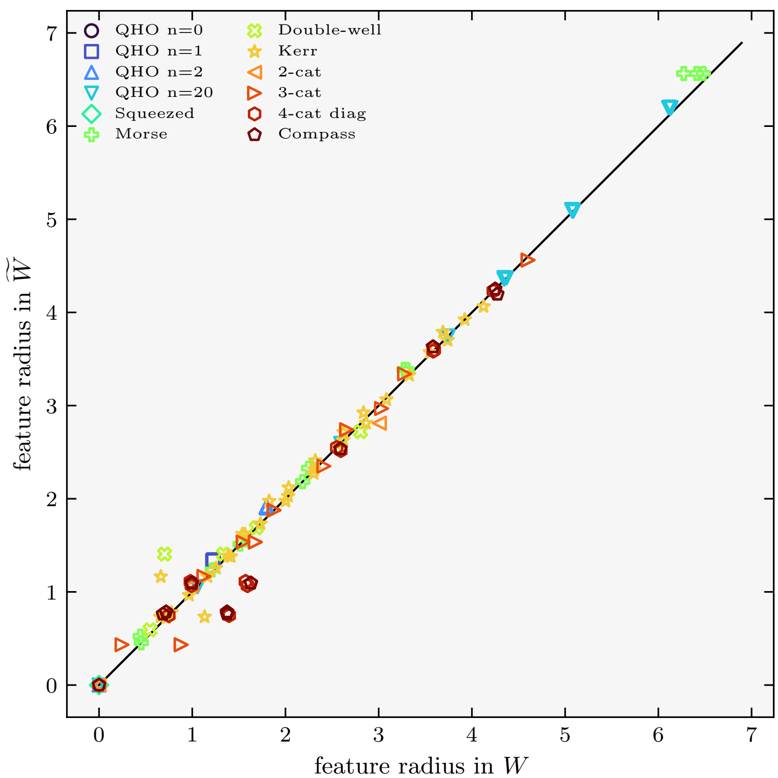

FIG. S1. Every positive feature of every state, plotted as its phase-space radius in W against its radius in $\widetilde{W}$. Each state carries one symbol and contributes one point per feature. The library falls on the diagonal, so the portrait places each feature at the radius the bare function gives it.

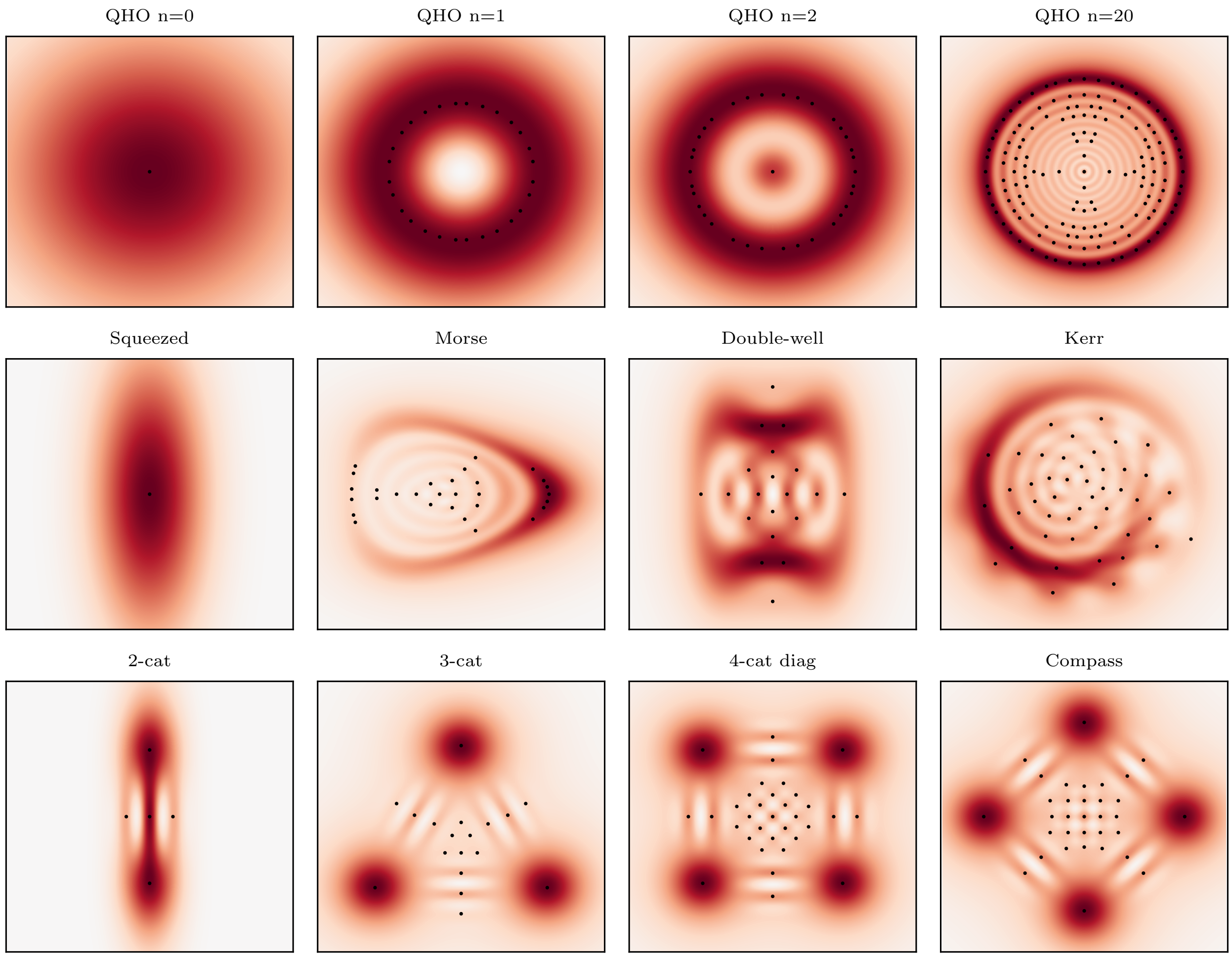

FIG. S2. The portrait $\widetilde{W}$ of each state as a heat map, with the positive maxima of the bare function W marked. The marks fall on the features of the portrait for every state, the rings of the Fock states, the lobes and interference peaks of the cats, and the curved structure of the anharmonic states.

S3.1. The transfer function matches the prediction

Section S2 reduces the portrait to a single convolution $\widetilde{W} = W * \bar{K}$ and gives the chord function of $\bar{K}$ in closed form, Eq. (S10). That chord function is the transfer that carries W to $\widetilde{W}$, isotropic in the scaled frame and cut off at the quorum cell $\tilde{a}$. This subsection measures it on every state.

In the scaled chord $|\xi|$ the prediction is a single curve set by the one ratio $\tilde{a}/(h/2)$ of the state. For each state the chord-domain ratio of $\widetilde{W}$ to W is formed as the power-weighted estimate $\mathrm{Re}\langle \widehat{W}^* \widehat{\widetilde{W}} \rangle / \langle |\widehat{W}|^2 \rangle$ in each chord bin and plotted against Eq. (S10) with no quantity fit. Figure S5 shows the twelve states. The vacuum and the squeezed state, at $\tilde{a} = h/2$, give a Gaussian, and the high-capacity states develop the soft edge of Eq. (S11), carrying structure out to $\tilde{a}$.

The angle-averaged kernel reproduces Eq. (S10) to better than 4×10^{-5} across the library, taken

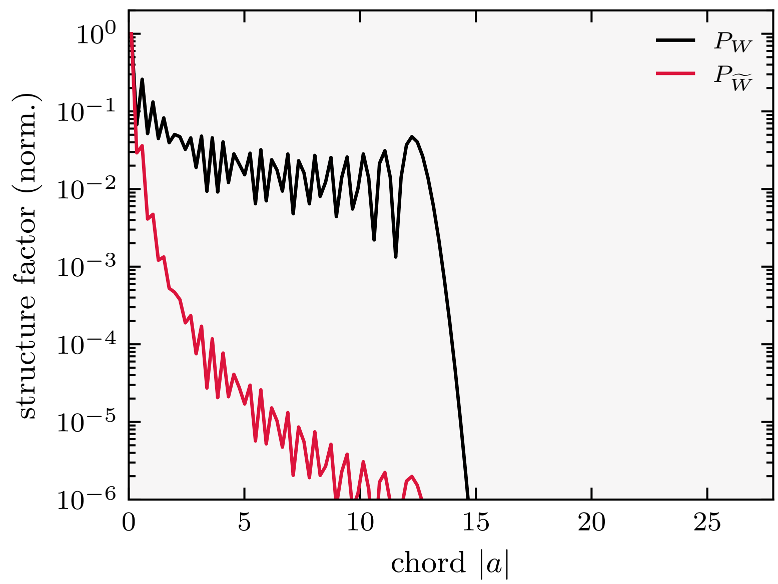

FIG. S3. Radially averaged chord power of W and $\widetilde{W}$ for the Fock state $n = 20$. The two functions occupy the same chords. The figure is read for where the curves carry power, not for how high they stand, since depth is not what this section measures.

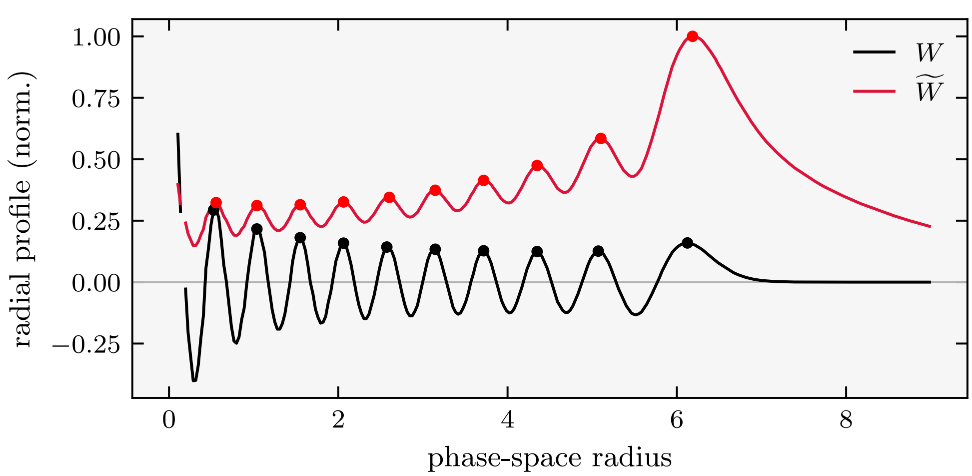

FIG. S4. Radial profiles of W and $\widetilde{W}$ for the Fock state $n = 20$. The positive rings sit at the same radii in both. The portrait rests above zero on its envelope while the bare function dips negative between rings, and neither difference moves a ring.

as the largest difference over the resolved chord disk, so the closed form is exact at the level of the numerics. On the states themselves the measured transfer tracks the same curve to a chord-band root-mean-square near one percent, the residual set by the radial binning rather than by a departure from the prediction.

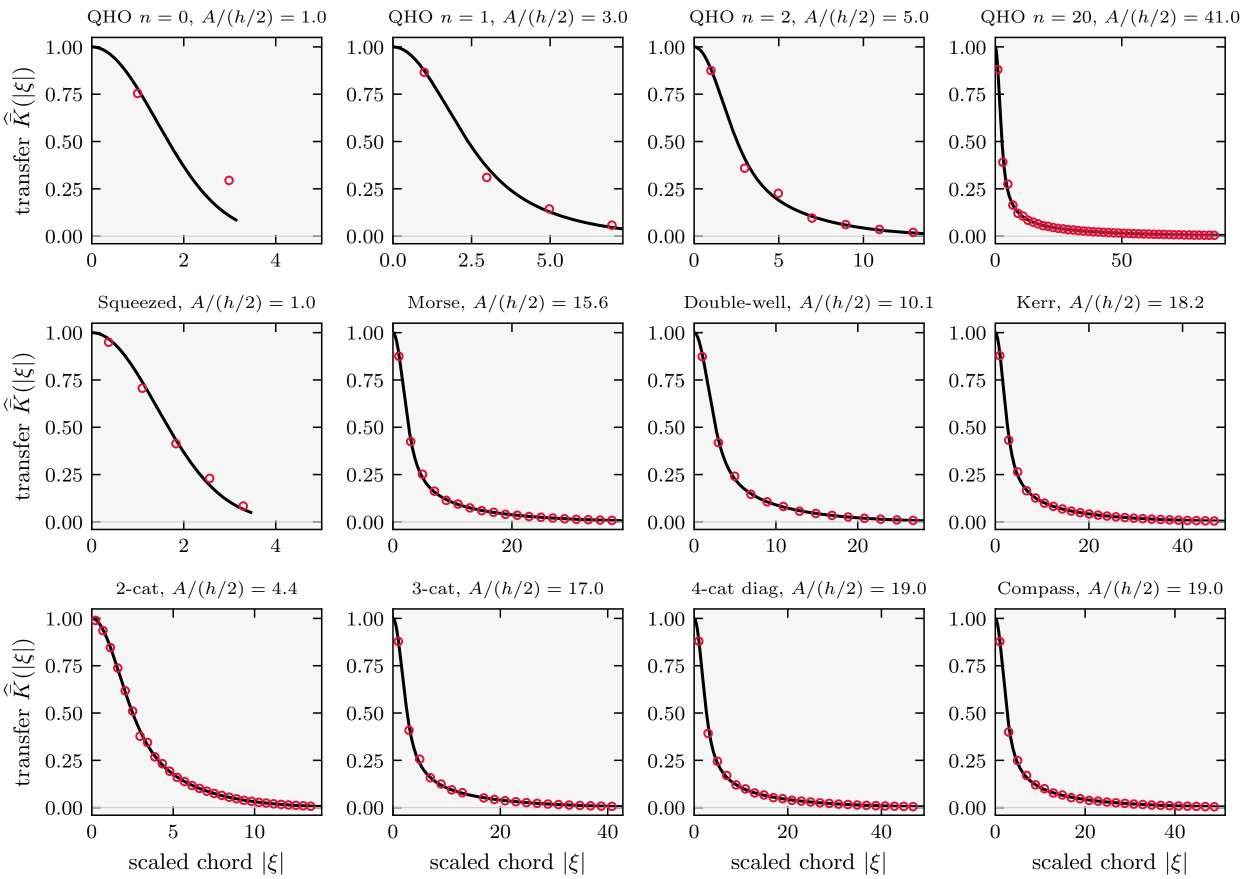

FIG. S5. Transfer function of the portrait across the twelve states. For each state the chord-domain ratio of $\widetilde{W}$ to W, measured on the scaled chord $|\xi|$ where Sec. S2 predicts an isotropic multiplier, is plotted as points against the curve of Eq. (S10) with the state's own ratio $\tilde{a}/(h/2)$. No quantity is fit. The vacuum and the squeezed state, at $\tilde{a} = h/2$, give a Gaussian, and the high-capacity states develop the soft edge of Eq. (S11) that carries structure out to the quorum cell $\tilde{a}$.

S3.2. The portrait carries the structure at the same positions

The cross-spectral phase between W and $\widetilde{W}$ is the standard test for a displacement of structure. A feature moved to a new position registers as a nonzero phase at the corresponding chord. Table S3 reports the in-band phase and the magnitude-squared coherence γ^2 for the twelve states. The phase is zero to machine precision for every state, so no feature moves. The coherence stays near unity. It falls below unity only through the radial average over a transfer that is mildly anisotropic, and the phase rather than the coherence carries the statement of position.

The radially averaged chord power removes the choice of axis. Convolution is multiplication in the chord domain, so W and $\widetilde{W}$ carry power at the same chords wherever the kernel transfer is free of zeros, which holds within the band. Figure S3 overlays the two spectra for the Fock state

$n = 20$. They occupy the same chords, the portrait lowered in amplitude and the bare function reaching further only into the fine scale below $\tilde{a}$ where its negativity lives. The rings of the $n = 20$ state, the finest structure in the library, sit at a spacing of $3.6\,\delta$ measured the same way in W and in $\widetilde{W}$, their radii listed in Table S5 and shown in Fig. S4.

The positive features of W and $\widetilde{W}$ occupy the same points of phase space. Detecting the positive maxima of each function with one rule, Table S4 reports the count found in each, the number matched, and the median displacement of a matched pair in units of the quorum semi-axis δ. Every state but the Fock state $n = 20$ places its matched features within a third of δ, and the $n = 20$ value of $1.2\,\delta$ is the coarse-grid sampling of its many rings rather than a displacement, tightening on the fine grid of Sec. S3 S3.4. The portrait carries fewer maxima than the bare function for the high-capacity states, because the shallowest features merge into the envelope rather than move. Figure S2 shows the portrait of each state with the positive maxima of the bare function marked, and Fig. S1 plots the phase-space radius of every matched feature in W against its radius in $\widetilde{W}$, where the whole library falls on the diagonal.

S3.3. Non-negativity

Convolving W with a phase-space Gaussian of action at least $h/2$ returns a non-negative function [5, 8]. The family-averaged kernel is a convex average of such Gaussians, so $\widetilde{W} \geq 0$ for every state. Table S6 reports the grid minimum of W and of $\widetilde{W}$, each relative to its own peak. The bare function runs to a depth comparable to its peak, and the portrait rests near zero, between 10^{-7} and the floating-point limit. This is the one property that separates the two functions.

S3.4. Method

Each portrait is the average of $W * K_\theta$ over 180 angles evenly spaced on $[0, \pi)$, evaluated on the uniform grid that carries W. The Fock state $n = 20$ is evaluated on a refined grid of 1001 points per axis, which resolves its rings, and the other states on the standard grid. The positive maxima of a function are its local maxima above five percent of its grid peak, found with a sliding-window maximum filter and matched to the nearest maximum of the other function within three quorum semi-axes. The coherence and phase are formed from the two-dimensional symplectic Fourier transforms of W and $\widetilde{W}$, binned radially in chord, with the in-band region taken where the power of W exceeds one percent of its peak. The grid reproduces the single-convolution identity

$\widetilde{W} = W * \bar{K}$ to 5×10^{-17}. The code reproducing every table and figure of this section is openly available [11].

state	$A/(h/2)$	γ^2	max $\lvert$phase$\rvert$
QHO $n = 0$	1.00	0.921	9×10^{-16}
QHO $n = 1$	3.00	0.947	6×10^{-16}
QHO $n = 2$	5.00	0.973	2×10^{-15}
QHO $n = 20$	41.00	0.998	1×10^{-15}
squeezed	1.00	0.944	1×10^{-15}
Morse	15.60	0.964	6×10^{-15}
double-well	10.14	0.938	6×10^{-16}
Kerr	18.21	0.994	3×10^{-15}
2-cat	4.35	0.800	7×10^{-16}
3-cat	17.00	0.992	4×10^{-15}
4-cat diag	19.00	0.995	6×10^{-16}
compass	19.00	0.995	1×10^{-14}

TABLE S3. Positions coincide. Magnitude-squared coherence γ^2 and the largest in-band cross-spectral phase between W and $\widetilde{W}$, in radians. The phase is zero to machine precision, so no feature is displaced. The coherence sits near unity, lowest for the 2-cat where the two-lobe transfer is most anisotropic.

state	$A/(h/2)$	N_W	$N_{\widetilde{W}}$	matched	offset/δ
QHO $n = 0$	1.00	1	1	1	0.00
QHO $n = 1$	3.00	28	40	28	0.23
QHO $n = 2$	5.00	35	41	35	0.29
QHO $n = 20$	41.00	149	97	87	1.23
squeezed	1.00	1	1	1	0.00
Morse	15.60	30	11	16	0.32
double-well	10.14	20	12	14	0.29
Kerr	18.21	46	31	32	0.33
2-cat	4.35	5	5	5	0.16
3-cat	17.00	18	14	17	0.39
4-cat diag	19.00	33	21	33	0.33
compass	19.00	33	21	29	0.33

TABLE S4. Positive features co-locate. Number of positive maxima found in W and in $\widetilde{W}$ by one detector, the number matched between them, and the median displacement of a matched pair in units of the quorum semi-axis δ. Matched features sit within a third of δ for every state but $n = 20$, whose $1.2\,\delta$ is coarse-grid sampling of its rings. The portrait shows fewer maxima where the shallowest features merge into the envelope.

ring	1	2	3	4	5
W	0.53	1.04	1.55	2.06	2.57
$\widetilde{W}$	0.56	1.04	1.55	2.06	2.60
ring	6	7	8	9	10
W	3.15	3.72	4.35	5.07	6.13
$\widetilde{W}$	3.15	3.72	4.35	5.10	6.19

TABLE S5. Positive-ring radii of the Fock state $n = 20$, the finest structure in the library. The rings of W and $\widetilde{W}$ coincide ring for ring, at a spacing of $3.6\,\delta$.

state	$A/(h/2)$	$\min W/\max W$	$\min \widetilde{W}/\max \widetilde{W}$
QHO $n = 0$	1.00	0.000	7×10^{-6}
QHO $n = 1$	3.00	−2.241	1×10^{-16}
QHO $n = 2$	5.00	−0.414	1×10^{-7}
QHO $n = 20$	41.00	−0.402	5×10^{-8}
squeezed	1.00	−0.000	-4×10^{-17}
Morse	15.60	−1.167	6×10^{-9}
double-well	10.14	−1.381	1×10^{-12}
Kerr	18.21	−1.002	8×10^{-9}
2-cat	4.35	−0.750	-6×10^{-17}
3-cat	17.00	−0.995	2×10^{-9}
4-cat diag	19.00	−0.760	1×10^{-7}
compass	19.00	−0.749	2×10^{-8}

TABLE S6. Non-negativity, the one asymmetry. Grid minimum of each function relative to its peak. The bare function runs to a depth comparable to its peak, the portrait rests near zero.

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